

REQUEST FOR PROPOSAL - DEMO PACKAGE

Hydrogen Process Control Valve Package

Hydrogen Purification and Export Header Control Valve Package

FICTIONAL CUSTOMER PROFILE

Helios Green Hydrogen and Ammonia Hub

INDUSTRY

Hydrogen / Energy Transition

ISSUE DATE

29 May 2026

BID DUE DATE

28 June 2026

COMMERCIAL CONTEXT

Estimated package value: INR 16.4 Cr

PACKAGE TYPE

Severe-service control valve proposal

Demo-safe fictional RFP package created for WinsProposal. Customer names are fictional industry-style accounts and do not represent official documents, endorsements, trademarks, or partnerships.

1. Purchaser Profile and Project Background

Helios Green Hydrogen and Ammonia Hub is a fictional energy-transition owner profile used to represent a renowned-style hydrogen project with strict material compatibility, traceability, safety, and documentation expectations.

The project requires hydrogen-compatible process control valves for purification, compression interface, and export header control. The purchaser requires evidence-based proposal responses for material compatibility, high-integrity sealing, leakage class, hazardous area accessories, documentation traceability, and engineering validation.

Proposal expectation: Supplier response shall show engineering proposal intelligence: extracted requirements, compliance/TBE readiness, severe-service risk recognition, knowledge reuse, deviations, review owners, and management-ready documentation.

2. Scope of Supply

- Supply hydrogen-compatible control valves and accessory packages.
- Provide material compatibility notes covering pressure boundary, trim, packing, gaskets, bolting, and accessories.
- Provide leakage class and high-integrity sealing proposal-stage narrative.
- Provide traceability dossier, inspection/testing plan, and compliance matrix.
- Provide clarification list for hydrogen composition, contaminants, pressure cases, area classification, and final leakage class.

3. Line Items

Tag / Ref.	Item	Qty	Indicative Size / Class	Service Notes
HV-H2-3101A/B/C/D	Hydrogen pressure control valve	4	NPS 3, ASME 600	Hydrogen-rich process gas
FV-H2-3150A/B	Hydrogen flow control valve	2	NPS 2, ASME 600	High-integrity sealing
PV-H2-3190	Export header control valve	1	NPS 4, ASME 900	Pressure letdown/noise review
DOC-H2	Traceability and compliance dossier	1 lot	MDR package	MTC, PMI, leakage, certificates

4. Process Conditions

Input	Requirement / Value
Fluid	Hydrogen-rich process gas
Inlet pressure	48 barg
Outlet pressure	18 barg
Temperature	60 C
Leakage class	Class V requested for selected tags
Service concerns	Hydrogen compatibility, high-integrity sealing, traceability, process safety
Area classification	Hazardous area certificates required where applicable

5. Technical Specification

- Supplier shall provide hydrogen compatibility narrative for all wetted and pressure-retaining components.
- Supplier shall state assumptions for embrittlement susceptibility, packing, gasket, bolting, and trim material selection.
- Supplier shall provide leakage class feasibility comments and identify tags requiring special sealing options.
- Supplier shall check gas pressure letdown, outlet velocity, choked-flow risk, and aerodynamic noise assumptions.
- Supplier shall provide hazardous area accessory certificate list where positioners, solenoids, or switches are included.
- Supplier shall clearly state final design and sizing require qualified engineering validation.

6. Compliance Requirements

- ISA 75.01 / IEC 60534 control valve sizing awareness.
- ASME B16.34 pressure-temperature considerations.
- Project hydrogen material compatibility specification.
- Project traceability, hazardous-area accessory, and documentation procedures.
- Client process safety and reliability requirements.

7. Inspection and Testing Requirements

- Pressure test per project basis.
- Seat leakage test by tag.
- PMI and material traceability verification.
- Packing/sealing documentation review.
- Accessory certificate review.
- Final traceability dossier.

8. Required Proposal-Stage Engineering Intelligence

Engineering Narrative Required

- Extract process conditions and classify application/service.
- Provide preliminary valve configuration narrative.
- Provide severe-service risk assessment.
- Provide cavitation/flashing/choked-flow narrative where relevant.
- Provide noise/vibration risk narrative.
- Provide trim, material, actuator, and accessory notes.

Calculation Summary Required

- State process inputs and assumptions.
- Classify pressure drop severity.
- Provide indicative Cv/Kv placeholder only where safe.
- List risk flags and missing data.
- List validation checklist and standards awareness.
- State that final sizing/design requires engineer validation.

9. Technical Bid Evaluation Template

Evaluation Tag	Purchaser Requirement
Material	Hydrogen-compatible material note required for all wetted components.
Trim	Trim to balance pressure letdown, leakage class, and hydrogen compatibility.
Sealing	High-integrity packing/gasket proposal required by tag.
Actuator	Actuator sized for fail action, shutoff pressure, leakage class, and safety requirements.
Accessories	Hazardous-area certificates required for positioner, solenoid, and switches.
Documentation	Traceability dossier, MTCs, PMI, leakage evidence, and certificates.

10. Evaluation Criteria

Criterion	Weight
Hydrogen compatibility	30%
Technical compliance	25%
Traceability and documentation	20%
Delivery	10%
Commercial value	15%

11. Possible Deviations and Clarifications

- Final material compatibility requires client material restriction list and hydrogen purity/contaminant data.
- Final leakage class claim requires confirmed shutoff pressure, temperature, trim design, and acceptance criteria.
- Noise guarantee requires final flow cases and outlet piping details.

12. Engineering Review Points

- Confirm hydrogen purity, contaminants, moisture, and operating envelope.
- Confirm material restriction list and embrittlement screening criteria.
- Confirm leakage class by tag and test acceptance standard.
- Confirm hazardous area classification and accessory certificate requirements.

- Confirm final traceability dossier format.

13. Commercial and Delivery Assumptions

- Supplier shall state drawing submission lead time, manufacturing lead time, test duration, and shipping readiness.
- Supplier shall identify long-lead materials, special trims, actuator accessories, and inspection hold points that may affect schedule.
- Supplier shall provide itemized pricing by line item and list optional compliance or testing costs separately.
- Supplier shall include a deviation register with technical, commercial, and schedule impact.
- Proposal validity shall be not less than 60 days from bid due date.

Mandatory wording: Any engineering analysis in the supplier proposal shall clearly state: "Preliminary proposal-stage engineering estimate. Final sizing/design must be validated by qualified engineers using company-approved tools and standards."